FACTORY IO- Sorting of Boxes(1) / PLC_1 [CPU 1212C AC/DC/Rly] / Program blocks

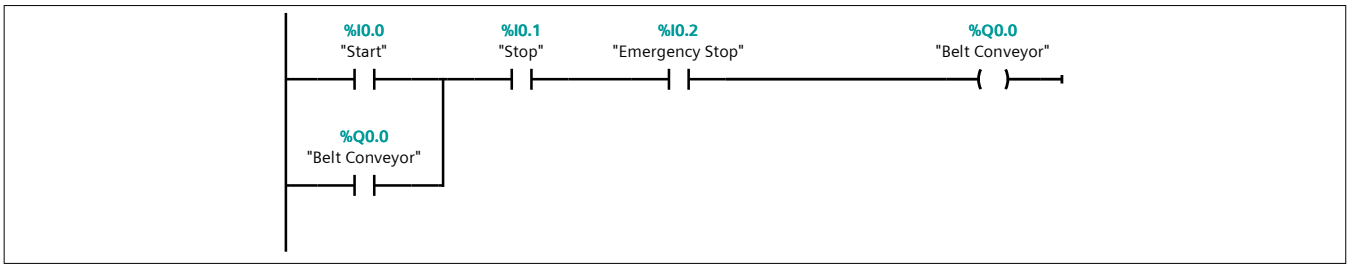
Main [OB1]

Main Properties					
General					
Name	Main	**Number**	1	**Type**	OB
Language	LAD	**Numbering**	automatic		
Information					
Title	NFI Siemens Course Exercise: Sorting of Boxes	**Author**	Rajvir	**Comment**	
Family	NFI	**Version**	0.1	**User-defined ID**	

Main			
Name	**Data type**	**Default value**	**Comment**
Input			
Initial_Call	Bool		Initial call of this OB
Remanence	Bool		=True, if remanent data are available
Temp			
Constant			

Network 1: Latching & Unlatching of conveyor Q0.0 from Start I0.0 and Stop I0.1

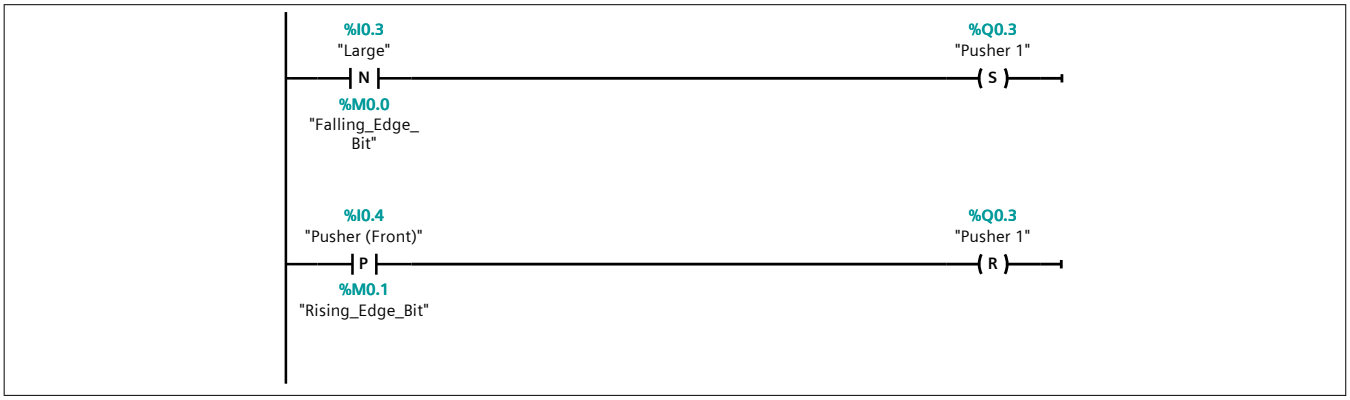
6. Use Emergency I0.2 to stop the conveyor Q0.0



Symbol	Address	Type	Comment
"Belt Conveyor"	%Q0.0	Bool	
"Emergency Stop"	%I0.2	Bool	
"Start"	%I0.0	Bool	Toggle Button
"Stop"	%I0.1	Bool	

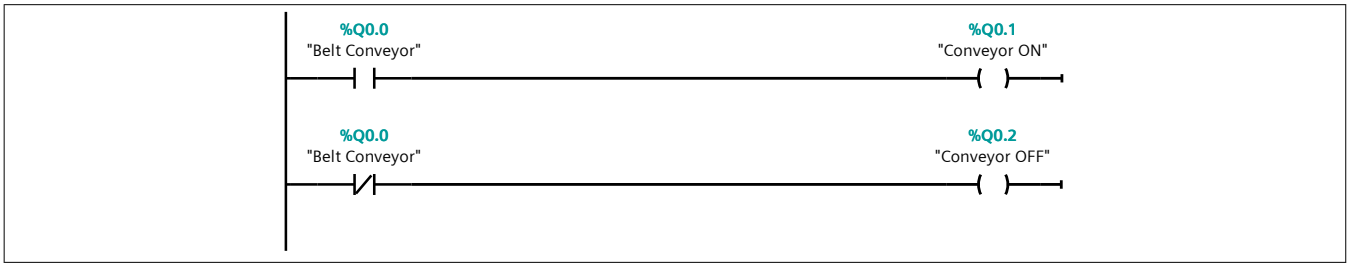
Network 2: Sense bigger box via Diffuse sensor I0.3 and Drag bigger boxes using Pusher Q0.3

4. Sense and control Pusher Q0.3 using Pusher front I0.4 and back limit I0.5



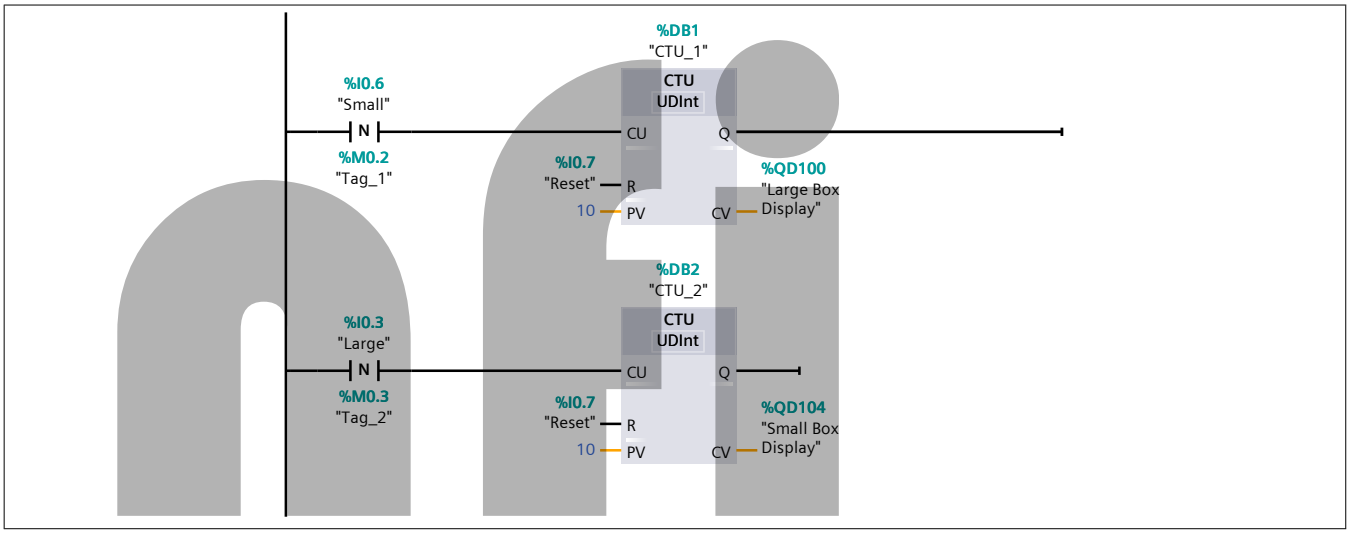
Symbol	Address	Type	Comment
"Falling_Edge_Bit"	%M0.0	Bool	
"Large"	%I0.3	Bool	
"Pusher 1"	%Q0.3	Bool	
"Pusher (Front)"	%I0.4	Bool	
"Rising_Edge_Bit"	%M0.1	Bool	

Network 3: 5. Indicate the Start & Stop operation of Conveyor via Indicators Q0.1 and Q0.2



Symbol	Address	Type	Comment
"Belt Conveyor"	%Q0.0	Bool	
"Conveyor OFF"	%Q0.2	Bool	
"Conveyor ON"	%Q0.1	Bool	

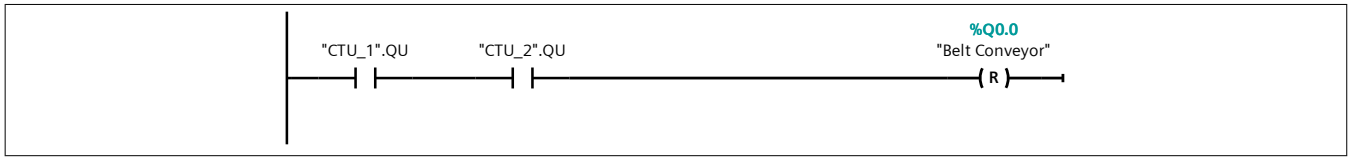
Network 4: Count big box via Diffuse sensor I0.3 and Count small box via Diffuse sensor I0.6



Symbol	Address	Type	Comment
"Large Box Display"	%QD100	UDInt	

Symbol	Address	Type	Comment
"Large"	%I0.3	Bool	
"Reset"	%I0.7	Bool	
"Small Box Display"	%QD104	UDInt	
"Small"	%I0.6	Bool	
"Tag_1"	%M0.2	Bool	
"Tag_2"	%M0.3	Bool	

Network 5: Stop the conveyor when we have at least 10 small and large boxes



Symbol	Address	Type	Comment
"Belt Conveyor"	%Q0.0	Bool	
"CTU_1".QU		Bool	
"CTU_2".QU		Bool	